

Assignment - 8

Calculate state value & Action value for each state.

sol The given policy is:-

$$\pi = \{s_0: \text{Right}, s_1: \text{Down}, s_2: \text{Right}, s_4: \text{UP}\}$$

$$\text{Move} = -1$$

$$\text{Goal} = +10$$

Therefore;

$$s_0 \xrightarrow{R} s_1$$

$$s_1 \xrightarrow{D} s_3$$

$$s_3 \xrightarrow{R} s_4$$

$$s_4 \xrightarrow{UP} \text{Goal}$$

$$s_0 \rightarrow s_1 \rightarrow s_3 \rightarrow s_4 \rightarrow \text{Goal}$$

$$-1 \quad -1 \quad -1 \quad 10$$

Calculate state value $V(s_0)$

$$V(s_0) = R_0 + \gamma R_1 + \gamma^2 R_2 + \gamma^3 R_3$$

$$R_0 = -1$$

$$R_1 = -1$$

$$R_2 = -1$$

$$R_3 = +10$$

$$\gamma = 0.9$$

$$\begin{aligned}
 V(s_0) &= (-1) + (0.9)(-1) + (0.9)^2(-1) + (0.9)^3(10) \\
 &= (-1) - 0.9 - 0.81 + 7.29 \\
 &= -2.71 + 7.29
 \end{aligned}$$

$$V(s_0) = 4.58$$

Calculate Action Value $Q(s_0, up)$

$$s_0 \rightarrow s_1 \rightarrow s_3 \rightarrow s_4 \rightarrow G$$

$$\begin{aligned}
 Q(s_0, up) &= -1 + \gamma V(s_0) \\
 &= -1 + (0.9)(4.58) \\
 &= -1 + 4.122
 \end{aligned}$$

$$Q(s_0, up) = 3.122$$

$$V(s_0) = 4.58$$

$$Q(s_0, up) = 3.122$$
